# Radiation by relativistic particles

So far we focused on monochromatic waves for which

$$\vec{B}(\vec{r},t) = \vec{B}(\vec{r})\, e^{-i\omega t}, \quad \vec{E}(\vec{r},t) = \vec{E}(\vec{r})\, e^{-i\omega t} \text{ and so on.}$$

But during collisions EM waves in a wide range of frequences are produced and we describe this now.

To do so, we implement Fourier decomposition

$$\cancel{\vec{B}(\vec{r},t) = \int \frac{d\omega}{2\pi} e^{+i\omega t} B} \qquad \vec{B}(\vec{r},t) = \int \frac{d\omega}{2\pi} e^{-i\omega t}\, \vec{B}(\vec{r},\omega)$$

$$\vec{B}(\vec{r},\omega) = \int dt\, e^{i\omega t}\, \vec{B}(\vec{r},t).$$

For monochromatic wave $\vec{B}'(\vec{r},\Omega) = \int dt\, e^{i\Omega t} \underbrace{\vec{B}(\vec{r})\, e^{-i\omega t}}_{\vec{B}(\vec{r},t)} =$

$$= \vec{B}(\vec{r}) \int dt\, e^{it(\Omega-\omega)}$$

$$= 2\pi\, \delta(\Omega-\omega)\, \vec{B}(\vec{r}).$$

Let's denote $\vec{B}(\vec{r},\Omega) \equiv \vec{B}_\Omega(\vec{r})$ : $\vec{B}_\Omega(\vec{r}) = 2\pi\, \delta(\Omega-\omega)\, \vec{B}(\vec{r})$.

In <u>lect 3A</u> we found $\left(k = \frac{\omega}{c}\right)$:

$$\vec{B}(\vec{r}) = \frac{i\omega}{c} \frac{\mu_0}{4\pi r} \int \hat{n} \times \vec{J}(\vec{r}')\, e^{i\frac{\omega}{c}r - i\frac{\omega}{c}\hat{n}\cdot\vec{r}'}\, d\vec{r}'$$

Multiply by $2\pi\, \delta(\Omega-\omega) \Rightarrow$ LHS becomes $\vec{B}_\Omega(\vec{r}) = 2\pi\, \delta(\Omega-\omega)\, \vec{B}(\vec{r})$

RHS involves $\vec{J}_\Omega(\vec{r}) = 2\pi\, \delta(\Omega-\omega)\, \vec{J}(\vec{r})$

$$\Rightarrow \vec{B}_\Omega(\vec{r}) = \frac{i\Omega}{c} \frac{\mu_0}{4\pi r} \int \hat{n} \times \vec{J}_\Omega(\vec{r}')\, e^{i\frac{\Omega}{c}r - i\frac{\Omega}{c}\hat{n}\cdot\vec{r}'}\, d\vec{r}'$$

By definition $\vec{J}_\Omega(\vec{r}') = \vec{J}(\vec{r}',\Omega) = \int_{-\infty}^{\infty} dt'\, e^{i\Omega t'}\, \vec{J}(\vec{r}',t')$

$$\Rightarrow \vec{B}_\Omega(\vec{r}) = \frac{i\Omega}{c} \frac{\mu_0}{4\pi r}\, e^{i\Omega r/c} \int d\vec{r}' \int_{-\infty}^{\infty} dt'\, e^{i\Omega t'}\, e^{-i\frac{\Omega}{c}\hat{n}\cdot\vec{r}'}\, \hat{n} \times \vec{J}(\vec{r}',t')$$

this is convenient for analyzing radiation of charged particles.

$$\vec{J}(\vec{r}\,',t') = q\,\vec{v}(t')\,\delta(\vec{r}\,' - \vec{r}_0(t'))$$

$\vec{r}_0(t)$ = particle's position

$\vec{v}(t) = \frac{d\vec{r}_0}{dt}$

change $\Omega \to \omega$

$$\vec{B}_\omega(\vec{r}) = \frac{i\omega}{c}\frac{\mu_0 q}{4\pi r} e^{i\frac{\omega}{c}r}\int_{-\infty}^{\infty} dt'\, e^{i\omega t'} e^{-i\frac{\omega}{c}\hat{n}\cdot\vec{r}_0(t')}\,\hat{n}\times\vec{v}(t')$$

Intensity $\vec{S}(\vec{r},t) = \frac{1}{\mu_0}\vec{E}(\vec{r},t)\times\vec{B}(\vec{r},t) = \frac{c}{\mu_0}\vec{B}^2(\vec{r},t)\,\hat{n}$

of emitted waves far from the particle, where waves become (locally) plane waves.

<u>Total</u> emitted energy into the area $r^2 d\Omega$

$$d\mathcal{E}_{rad} = \int_{-\infty}^{\infty} dt\; \vec{S}(\vec{r},t)\cdot\hat{n}\; r^2 d\Omega$$

$$\frac{d\mathcal{E}_{rad}}{d\Omega} = \frac{c r^2}{\mu_0}\int_{-\infty}^{\infty} dt\; \vec{B}^2(\vec{r},t) = \int_{-\infty}^{\infty} dt\,\frac{dP}{d\Omega}$$

$$\int dt\; \vec{B}^2(r,t) = \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty}\frac{d\omega_1}{2\pi} e^{-i\omega_1 t}\,\vec{B}(\vec{r},\omega_1)\cdot\int_{-\infty}^{\infty}\frac{d\omega_2}{2\pi} e^{-i\omega_2 t}\,\vec{B}(\vec{r},\omega_2)$$

$$= \left(\int_{-\infty}^{\infty} dt\; e^{-it(\omega_1+\omega_2)} = 2\pi\,\delta(\omega_1+\omega_2)\right) =$$

$$= \int_{-\infty}^{\infty}\frac{d\omega_1}{2\pi}\;\vec{B}(\vec{r},\omega_1)\cdot\vec{B}(\vec{r},-\omega_1)$$

But $\vec{B}(\vec{r},t) = \text{real} = \int\frac{d\omega}{2\pi} e^{-i\omega t}\,\vec{B}(\vec{r},\omega)$

$$\cancel{= \vec{B}^*(\vec{r},t) = \int\frac{d\omega}{2\pi} e^{i\omega t}\,\vec{B}^*(\vec{r},\omega) =}$$

$$\cancel{= \int\frac{d\omega}{2\pi} e^{i(-\omega)t}} \qquad = \vec{B}^*(\vec{r},\omega)$$

$$\Rightarrow \vec{B}(\vec{r},-\omega) = \int dt\; e^{i(-\omega)t}\,\vec{B}(\vec{r},t) = \left(\int dt\; e^{i\omega t}\,\vec{B}(\vec{r},t)\right)^* =$$

$$\Rightarrow \frac{d\mathcal{E}_{rad}}{d\Omega} = \frac{c r^2}{\mu_0} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \vec{B}(\vec{r},\omega) \cdot \vec{B}^*(\vec{r},\omega) =$$

$$= \frac{c r^2}{\mu_0} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} |\vec{B}(\vec{r},\omega)|^2 = \frac{2 c r^2}{\mu_0} \int_0^{\infty} \frac{d\omega}{2\pi} |\vec{B}(\vec{r},\omega)|^2 =$$

$$= \int_0^{\infty} d\omega \frac{d^2 \mathcal{E}_{rad}}{d\omega\, d\Omega}$$ ← this is the definition of the angular spectral fluence of the radiation

$$\Rightarrow \frac{d^2 \mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{2 c r^2}{2\pi\mu_0} |\vec{B}_\omega(\vec{r})|^2 = \frac{c r^2}{\pi \mu_0} \cdot$$

$$\cdot \left| \frac{i\omega}{c} \frac{\mu_0 q}{4\pi r} e^{i\frac{\omega r}{c}} \int_{-\infty}^{\infty} dt' \, e^{i\omega(t' - \hat{n}\cdot\vec{r}_0/c)} \hat{n} \times \vec{V}(t') \right|^2$$

$$= \frac{\mu_0 q^2 \omega^2}{16\pi^3 c} \left| \int_{-\infty}^{\infty} dt \, e^{i\omega(t - \hat{n}\cdot\vec{r}_0(t)/c)} \hat{n} \times \vec{V}(t) \right|^2$$

If we were to start with $\vec{S}(\vec{r},t) = \frac{1}{c\mu_0} \vec{E}^2(\vec{r},t)\,\hat{n}$, we would obtain

$$\frac{d^2 \mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{\mu_0 q^2 \omega^2}{16\pi^3 c} \left| \int dt \, e^{i\omega(t - \hat{n}\cdot\vec{r}_0(t)/c)} \hat{n} \times (\hat{n} \times \vec{V}(t)) \right|^2$$

but $|\hat{n} \times \vec{v}| = |\hat{n} \times (\hat{n} \times \vec{v})| \Rightarrow$ magnitude is the same

since $(\hat{n} \times \vec{v})^2 = v^2 \sin^2\theta$

$$(\hat{n} \times (\hat{n} \times \vec{v}))^2 = (\vec{v} - \hat{n}\,\hat{n}\cdot\vec{v})^2 = (\vec{v} - \hat{n}\, v\cos\theta)^2 =$$

$$= v^2 + v^2\cos^2\theta - 2v\cos\theta(\vec{v}\cdot\hat{n}) = v^2(1 + \cos^2\theta - 2\cos^2\theta)$$

let $\displaystyle\int_{-\infty}^{\infty} dt\; i\omega\left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right) e^{i\omega(t-\hat{n}\cdot\vec{r}_0/c)} \frac{\hat{n}\times(\hat{n}\times\vec{v})}{i\omega\left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right)} =$

$$= \int_{-\infty}^{\infty} dt\; \frac{d}{dt}\left(e^{i\omega(t-\hat{n}\cdot\vec{r}_0/c)}\right) \frac{\hat{n}\times\hat{n}\times\vec{v}}{i\omega\left(1-\hat{n}\cdot\vec{v}/c\right)}$$

$$= \frac{1}{i\omega}\, e^{i\omega(t-\hat{n}\cdot\vec{r}_0/c)} \frac{\hat{n}\times\hat{n}\times\vec{v}}{1-\hat{n}\cdot\vec{v}/c}\Bigg|_{-\infty}^{+\infty}$$

this term is zero because waves decay at $\pm\infty$ (via $\omega\to\omega+i\delta$)

$$- \frac{1}{i\omega}\int_{-\infty}^{\infty} dt\; e^{i\omega(t-\hat{n}\cdot\vec{r}_0/c)} \frac{d}{dt}\left(\frac{\hat{n}\times(\hat{n}\times\vec{v})}{1-\hat{n}\cdot\vec{v}/c}\right)$$

this can also be $\frac{d}{dt}\left(\frac{\hat{n}\times\vec{v}}{1-\hat{n}\cdot\frac{\vec{v}}{c}}\right)$

$\vec{a}=\frac{d\vec{v}}{dt}$

Next, $\displaystyle\frac{d}{dt}\,\frac{\hat{n}(\hat{n}\cdot\vec{v})-\vec{v}}{1-\hat{n}\cdot\vec{v}/c} = \frac{\hat{n}(\hat{n}\cdot\vec{a})-\vec{a}}{1-\hat{n}\cdot\vec{v}/c} - \frac{\hat{n}(\hat{n}\cdot\vec{v})-\vec{v}}{\left(1-\hat{n}\cdot\vec{v}/c\right)^2}\cdot$

$$\cdot\left(-\hat{n}\cdot\frac{\vec{a}}{c}\right) = \frac{\left(\hat{n}(\hat{n}\cdot\vec{a})-\vec{a}\right)\left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right)+\left(\hat{n}\cdot\frac{\vec{a}}{c}\right)\left[\hat{n}(\hat{n}\cdot\vec{v})-\vec{v}\right]}{\left(1-\hat{n}\cdot\vec{v}/c\right)^2}$$

$$= \Big(\hat{n}(\hat{n}\cdot\vec{a})-\vec{a} - \cancel{\hat{n}(\hat{n}\cdot\vec{a})\left(\hat{n}\cdot\frac{\vec{v}}{c}\right)} + \vec{a}\left(\hat{n}\cdot\frac{\vec{v}}{c}\right)$$

$$+ \cancel{\hat{n}(\hat{n}\cdot\vec{v})\left(\hat{n}\cdot\frac{\vec{a}}{c}\right)} - \vec{v}\left(\hat{n}\cdot\frac{\vec{a}}{c}\right)\Big)\frac{1}{\left(1-\hat{n}\cdot\vec{v}/c\right)^2}$$

$$= \frac{\left(\hat{n}-\frac{\vec{v}}{c}\right)\left(\hat{n}\cdot\vec{a}\right) - \left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right)\vec{a}}{\left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right)^2} =$$

$$= \frac{\hat{n}\times\left(\hat{n}-\frac{\vec{v}}{c}\right)\times\vec{a}}{\left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right)^2}$$

$$\Rightarrow \frac{d^2\mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{\mu_0 q^2\omega^2}{16\pi^3 c}\left|\frac{i}{\omega}\int_{-\infty}^{\infty} dt\; e^{i\omega(t-\hat{n}\cdot\vec{r}_0/c)}\frac{\hat{n}\times\left(\hat{n}-\frac{\vec{v}}{c}\right)\times\vec{a}}{\left(1-\hat{n}\cdot\frac{\vec{v}}{c}\right)^2}\right|^2$$

# Scattering of relativistic particles and radiation it produces.

Brau 10.2

Radiation produced in the collision is called "bremsstrahlung" (german for "braking radiation")

Need to distinguish "hard" and "soft" collisions

$$\text{"hard"} \Rightarrow \omega\tau \ll 1 \qquad \text{"soft"} \Rightarrow \omega\tau \gg 1$$

where $\omega$ = radiation frequency
$\tau$ = duration of the collision

## Hard

$$\frac{d^2 \mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c} \left| \int_{-\infty}^{\infty} dt\, e^{i\omega(t - \hat{n}\cdot\vec{r}_0/c)} \frac{d}{dt}\left(\frac{\hat{n}\times\vec{v}}{1-\hat{n}\cdot\frac{\vec{v}}{c}}\right)\right|^2$$

In "hard" collisions $\int dt \sim \tau$ since before/after the collision velocity of the particle does not change

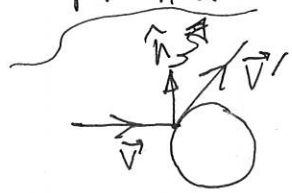
$\Rightarrow$ $e^{i\omega(t-\hat{n}\cdot\vec{r}_0/c)} \approx e^{i\omega\tau}$ can be taken outside the integral

$$\Rightarrow \frac{d^2 \mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c} \left| \frac{\hat{n}\times\vec{v}}{1-\hat{n}\cdot\frac{\vec{v}}{c}} \Bigg|_{-\infty}^{+\infty} \right|^2 = \begin{pmatrix} \vec{v}' = \vec{v}(t=+\infty) \\ \vec{v} = \vec{v}(t=-\infty) \end{pmatrix}$$

$$= \frac{\mu_0 q^2}{16\pi^3 c} \left| \frac{\hat{n}\times\vec{v}'}{1-\hat{n}\cdot\vec{v}'/c} - \frac{\hat{n}\times\vec{v}}{1-\hat{n}\cdot\vec{v}/c} \right|^2.$$

Relativistic motion $v \approx c$

$\Rightarrow \frac{1}{1-\hat{n}\cdot\frac{\vec{v}}{c}}$ is peaked for $\hat{n} \parallel \vec{v}$.




Non-relativistic motion $v \ll c$:

$$\frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c} \left|\hat{n}\times(\vec{v}' - \vec{v})\right|^2.$$

$\Rightarrow$ like dipole along $(\vec{v}' - \vec{v})$.

* Can apply this to $\beta$-decay neutron $\rightarrow$ proton + electron + neutrino

$q=0$ $\quad$ $q=+e$ $\quad$ $q=-e$

$\Rightarrow$ Instantaneous emission of charged particle

$\Rightarrow \vec{v} = \vec{v}_{initial} = 0 \quad , \vec{v}' = \vec{v}_{final} = \vec{v}$

$$\frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 e^2}{16\pi^3 c}\left(\frac{\hat{n}\times\vec{v}}{1-\frac{1}{c}\hat{n}\cdot\vec{v}}\right)^2 = \frac{\mu_0 e^2}{16\pi^3 c}\,\frac{v^2\sin^2\theta}{\left(1-\frac{v}{c}\cos\theta\right)^2}$$

$$\Rightarrow \frac{d\mathcal{E}}{d\omega} = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta\, \frac{1-\cos^2\theta}{\left(1-\frac{v}{c}\cos\theta\right)^2}\,\frac{\mu_0 e^2 v^2}{16\pi^3 c} = \quad (x=\cos\theta)$$

$$= \frac{\mu_0 e^2 v^2}{8\pi^2 c}\int_{-1}^{1} dx\, \frac{1-x^2}{\left(1-\frac{v}{c}x\right)^2}$$

$$I = \frac{c}{v}\int_{-1}^{1} d\left[\frac{1}{\left(1-\frac{v}{c}x\right)}\right](1-x^2) = \frac{c}{v}\left\{\frac{1-x^2}{1-\frac{v}{c}x}\Bigg|_{-1}^{1} + \int_{-1}^{1} dx\, \frac{2x}{1-\frac{v}{c}x}\right\} =$$

$$= \frac{c}{v}\left\{\frac{1}{1-\frac{v}{c}} - \frac{1}{1+\frac{v}{c}} - \frac{2c}{v}\int_{-1}^{1} d\left[\ln\left(1-\frac{v}{c}x\right)\right]x\right\}$$

$$= \frac{c}{v}\left\{\frac{2\frac{v}{c}}{1-\left(\frac{v}{c}\right)^2} - \frac{2c}{v}\left[x\ln\left(1-\frac{v}{c}x\right)\Bigg|_{-1}^{1} - \int_{-1}^{1} dx\, \ln\left(1-\frac{v}{c}x\right)\right]\right\}$$

$$y = \frac{v}{c}x$$

$$= \frac{2}{1-\left(\frac{v}{c}\right)^2} - 2\left(\frac{c}{v}\right)^2\left\{\ln\left(\frac{1-\frac{v}{c}}{1+\frac{v}{c}}\right) - \frac{c}{v}\underbrace{\int_{-v/c}^{v/c} dy\, \ln(1-y)}_{\left[-y-(1-y)\ln(1-y)\right]\Big|_{-v/c}^{v/c}}\right\}$$

$$\ln\left(1-\frac{v^2}{c^2}\right)$$

$$\int_{-1}^{1} dx \, \frac{1-x^2}{(1-\beta x)^2} = \frac{1}{\beta}\int_{-1}^{1} d\left(\frac{1}{1-\beta x}\right)(1-x^2) = \frac{1}{\beta}\left\{ \frac{1-x^2}{1-\beta x}\Big|_{-1}^{1} - (-2)\int_{-1}^{1} dx \, \frac{x}{1-\beta x} \right\}$$

$$= \frac{2}{\beta}\int_{-1}^{1} dx \, \frac{1-\beta x - 1}{1-\beta x}\left(-\frac{1}{\beta}\right) = -\frac{2}{\beta^2}\int_{-1}^{1} dx \left(1 - \frac{1}{1-\beta x}\right) =$$

$$= -\frac{4}{\beta^2} + \frac{2}{\beta^3}\int_{-\beta}^{\beta} \frac{d(\beta x)}{1-\beta x} = -\frac{4}{\beta^2} - \frac{2}{\beta^3}\ln(1-x)\Big|_{-\beta}^{\beta} =$$

$$= -\frac{4}{\beta^2} - \frac{2}{\beta^3}\ln\frac{1-\beta}{1+\beta} = \frac{2}{\beta^3}\left(\ln\frac{1+\beta}{1-\beta} - 2\beta\right) \qquad \beta = \frac{v}{c}$$

$$\Rightarrow \frac{d\mathcal{E}}{d\omega} = \frac{\mu_0 e^2 v^2}{8\pi^2 c}\,\frac{2c^3}{v^3}\left[\ln\left(\frac{c+v}{c-v}\right) - \frac{2v}{c}\right] =$$

$$= \frac{\mu_0 e^2 c}{4\pi^2}\left[\frac{c}{v}\ln\left(\frac{c+v}{c-v}\right) - 2\right] \rightarrow \begin{cases} \frac{2}{3}\left(\frac{v}{c}\right)^2 & \text{for } v \ll c \\ \ln\left(\frac{2c}{c-v}\right) & \text{for } v \to c \end{cases}$$

Observe $\frac{d\mathcal{E}}{d\omega}$ is $\omega$-independent

$\Rightarrow$ total energy radiated $\mathcal{E} = \int \frac{d\mathcal{E}}{d\omega}\, d\omega \rightarrow \infty$

which is result of our approximating $\beta$-decay as an instantaneous process.

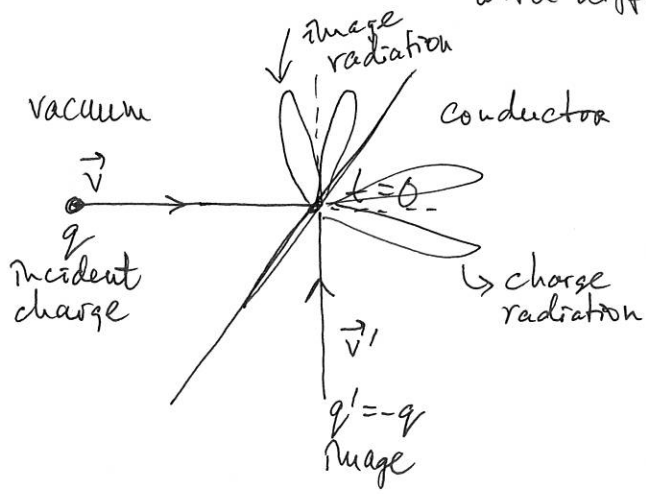
This implies infinite acceleration $\Rightarrow$ divergent radiated energy.

The integral $\int d\omega$ has to be bounded by

$\hbar\omega_{max} = \frac{1}{2}mv^2$ (which comes from Quantum mechanics) 10.2.2

<u>Transition radiation</u>: radiation of charged particle that crosses interface between two regions with different indices of refraction.

$t=-\infty$




When charged particle reaches conductor's surface, the charge $q$ and its image $q'$ cancel out → example of instantaneous charge disappearance ⇒ transition radiation.

Recall $$\frac{d^2\mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{2cr^2}{2\pi\mu_0}\left|\vec{B}_\omega(\vec{r})\right|^2$$

↳ total field $= \sum_j \vec{B}_\omega(\vec{r}_j)$

$$\Rightarrow \frac{d^2\mathcal{E}_{rad}}{d\omega\, d\Omega} = \frac{\mu_0}{16\pi^3 c}\left|\sum_j q_j \int_{-\infty}^{0} dt\, e^{i\omega(t-\hat{n}\cdot\vec{r}_j/c)}\frac{d}{dt}\left(\frac{\hat{n}\times\vec{v}_j}{1-\vec{n}\cdot\vec{v}_j/c}\right)\right|^2$$

$$q_j = \begin{cases} q & \text{for original charge} \\ -q & \text{for its image} \end{cases} ; \quad \vec{v}_j = \begin{cases} \vec{v} & \text{for } q. \\ \vec{v}' & \text{for } q'=-q. \end{cases}$$

Observe that $$\vec{v}_j(t) = \begin{cases} \text{const} = \vec{v}_j & \text{for } -\infty < t < 0 \\ 0 & \text{for } t \geq 0 \text{ (charge disappears)} \end{cases}$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\hat{n}\times\vec{v}_j}{1-\frac{1}{c}\hat{n}\cdot\vec{v}_j}\right) \neq 0$$ only during short "collision" interval when $t \approx 0$.

$$\Rightarrow \frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c}\left|\frac{\hat{n}\times\vec{v}}{1-\frac{1}{c}\hat{n}\cdot\vec{v}} - \frac{\hat{n}\times\vec{v}'}{1-\frac{1}{c}\hat{n}\cdot\vec{v}'}\right|^2$$

Alternatively, we can start with

$$\frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 \omega^2}{16\pi^3 c}\left|\sum_j \int_{-\infty}^{0} dt\; e^{i\omega(t-\hat{n}\cdot\vec{r}_j/c)}\; q_j\, \hat{n}\times\vec{v}_j\right|^2$$

but $\vec{r}_j = \vec{v}_j t \Rightarrow \int_{-\infty}^{0} dt\; e^{i\omega(t-\hat{n}\cdot\vec{v}_j/c\, t)} =$

$$= \int_{-\infty}^{0} dt\; e^{it(\omega - i\delta)(1-\frac{1}{c}\hat{n}\cdot\vec{v}_j)} = \left.\frac{e^{it(\omega-i\delta)(1-\frac{1}{c}\hat{n}\cdot\vec{v}_j)}}{i(\omega-i\delta)(1-\frac{1}{c}\hat{n}\cdot\vec{v}_j)}\right|_{-\infty}^{0}$$

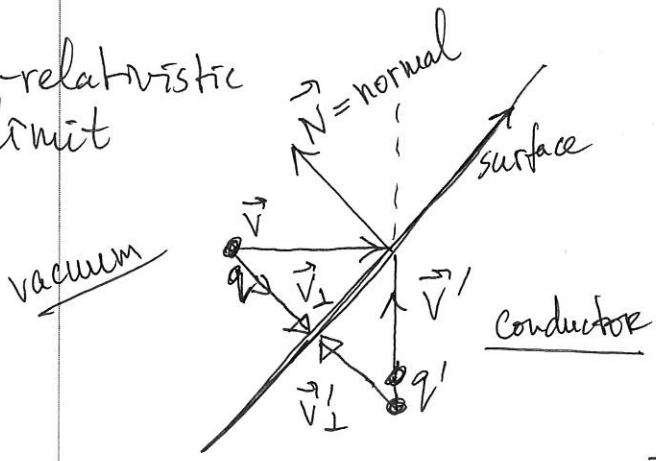
convergence factor

$$= \frac{1}{i\omega(1-\frac{1}{c}\hat{n}\cdot\vec{v}_j)}$$

$$\Rightarrow \frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0}{16\pi^3 c}\left|\sum_j \frac{q_j\, \hat{n}\times\vec{v}_j}{1-\frac{1}{c}\hat{n}\cdot\vec{v}_j}\right|^2$$

which is the same result.

Non-relativistic limit




$$\vec{v} = \vec{v}_{\parallel} + \vec{v}_{\perp} = \vec{v}_{\parallel} - \vec{N}(\vec{v}\cdot\vec{N})$$

parallel to the interface

normal to the interface

$$\vec{v}\,' = \vec{v}\,'_{\parallel} + \vec{v}\,'_{\perp} = \vec{v}_{\parallel} + \vec{N}(\vec{v}\cdot\vec{N})$$

$$\Rightarrow \vec{v}\,' - \vec{v} = 2\vec{N}(\vec{v}\cdot\vec{N})$$

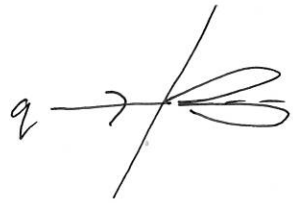
$$\Rightarrow \frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c}\left|\hat{n}\times(\vec{v}-\vec{v}\,')\right|^2 = \frac{\mu_0 q^2}{4\pi^3 c}(\vec{v}\cdot\vec{N})^2\left|\hat{n}\times\vec{N}\right|^2$$

$\Rightarrow$ radiation from the dipole oriented along the normal $\vec{N}$ to the conductor at the point of "contact" with charge $q$.

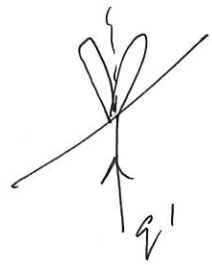
Relativistic limit: denominator $\dfrac{1}{1-\frac{1}{c}\hat{n}\cdot\vec{v}_j}$

is strongly peaked at $\hat{n} \parallel \vec{v}_j$

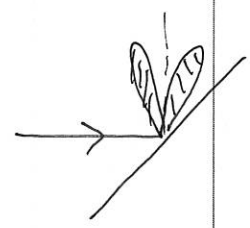
$\Rightarrow$ incident charge emits mostly <u>into</u> the conductor




while image charge $q'$ emits mostly into the vacuum region (where the observed radiation is)




$\Rightarrow$ we observe



$$\Rightarrow \frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c}\left|\frac{\hat{n}\times\vec{v}'}{1-\frac{1}{c}\hat{n}\cdot\vec{v}'}\right|^2$$

$$\hat{n}\cdot\vec{v}' = v'\cos\theta \approx v'\left(1-\tfrac{1}{2}\theta^2\right)$$

$$\hat{n}\times\vec{v}' = v'\sin\theta \approx v'\theta$$

$$\frac{v'\theta}{1-\frac{v'}{c}\left(1-\frac{1}{2}\theta^2\right)} = (v'=v) = \frac{v\theta}{1-\frac{v}{c}+\frac{v}{2c}\theta^2} = (v\approx c)$$

$$= \frac{c\theta}{\left(1-\frac{v}{c}\right)\left(1+\frac{\theta^2}{2\left(1-\frac{v}{c}\right)}\right)} = \frac{2\gamma^2 c\theta}{1+\gamma^2\theta^2}$$

But $\frac{1}{\gamma} = \sqrt{1-\frac{v^2}{c^2}} = \sqrt{\left(1+\frac{v}{c}\right)\left(1-\frac{v}{c}\right)} \approx \sqrt{2\left(1-\frac{v}{c}\right)}$ for $v\approx c$

$$\Rightarrow 1-\frac{v}{c} \approx \frac{1}{2\gamma^2}$$

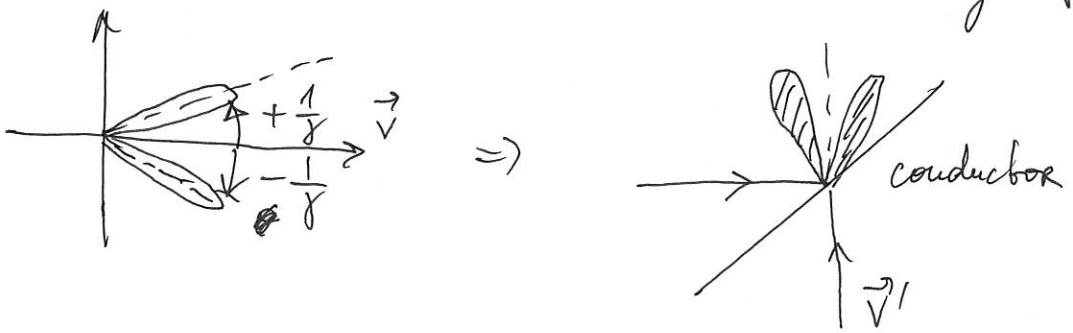
$$\frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2}{16\pi^3 c}\left(\frac{2\gamma^2 c\ \theta}{1+\gamma^2\theta^2}\right)^2 = \frac{\mu_0 c q^2}{4\pi^3}\frac{\theta^2}{(1+\gamma^2\theta^2)^2}$$

Analyze function $\dfrac{\theta^2}{(1+\gamma^2\theta^2)^2} = f(\theta^2)$

$$f(x) = \frac{x}{(1+\gamma^2 x)^2} \Rightarrow f'(x) = \frac{1}{(1+\gamma^2 x)^2} - \frac{2x\gamma^2}{(1+\gamma^2 x)^3} =$$

$$= \frac{1+\gamma^2 x - 2\gamma^2 x}{(1+\gamma^2 x)^3} = \frac{1-\gamma^2 x_0}{(1+\gamma^2 x)^3} = 0$$

$\Rightarrow x_0 = \theta_0^2 = \dfrac{1}{\gamma^2} \Rightarrow$ it is peaked at $\pm\theta_0 = \dfrac{1}{\gamma} = \sqrt{1-\dfrac{v^2}{c^2}}$



Coherent emission from multiple particles.

$$\frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2 \omega^2}{16\pi^3 c} \left| \sum_{j=1}^{N} \int_{-\infty}^{\infty} dt\, e^{i(\omega t - \vec{k}\cdot\vec{r}_j(t))}\, \hat{n}\times\vec{v}_j \right|^2$$

for a system of N particles (with $q_j = q$.)

Here $\vec{k} = \frac{\omega}{c}\hat{n}$ so that $\vec{k}\cdot\vec{r}_j = \omega\,\hat{n}\cdot\vec{r}_j/c$ as before.

If particles motion is not correlated, then individual integrals in the sum $\sum_j$ will come in with different random phases $e^{i\vec{k}\cdot\vec{r}_j}$ $\Rightarrow$ no interference

$$\Rightarrow \frac{d^2\mathcal{E}}{d\omega\, d\Omega} \simeq \frac{\mu_0 q^2 \omega^2}{16\pi^3 c} \sum_j \left| \int dt\, e^{i(\omega t - \vec{k}\cdot\vec{r}_j)}\, \hat{n}\times\vec{v}_j \right|^2$$

= sum of individual contributions.

However, if particle velocities <u>are correlated</u>

$$\vec{r}_j(t) = \vec{r}_0(t) + \vec{R}_j$$

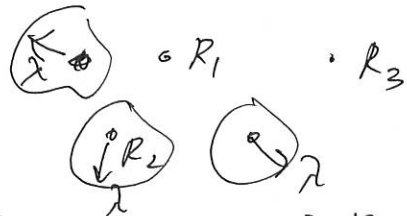
$\vec{R}_j$ ← random initial positions

$\vec{r}_0(t)$ ↑ same velocity

$$\Rightarrow \frac{d^2\mathcal{E}}{d\omega\, d\Omega} = \frac{\mu_0 q^2 \omega^2}{16\pi^3 c} \left| \sum_{j=1}^{N} e^{-i\vec{k}\cdot\vec{R}_j} \int_{-\infty}^{\infty} dt\, e^{i(\omega t - \vec{k}\cdot\vec{r}_0)}\, \hat{n}\times\vec{v}_0 \right|^2$$

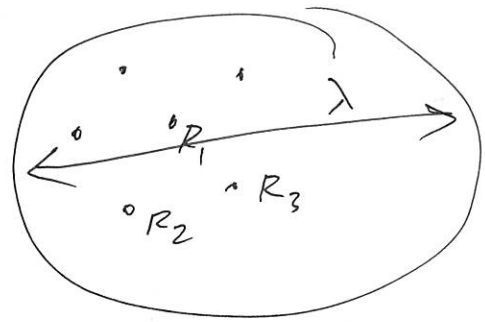
$$\left| \sum_j e^{-i\vec{k}\cdot\vec{R}_j} \right|^2 = \left| \sum_j e^{-i 2\pi \vec{R}_j\cdot\hat{n}/\lambda} \right|^2$$

If $\vec{k}\cdot\vec{R}_j \gg 1$




then again individual phases are random $\Rightarrow \left| \sum_j e^{-i\vec{k}\vec{R}_j} \right|^2 \approx \sum_j \left| e^{-i\vec{k}\cdot\vec{R}_j} \right|^2 = N.$

However if $\vec{k}\cdot\vec{R}_j \ll 1$ so that $\frac{R_j}{\lambda} \ll 1$




then $e^{-i\vec{k}\cdot\vec{R}_j} \approx 1$

$$\Rightarrow \left|\sum_j e^{-i\vec{k}\cdot\vec{R}_j}\right|^2 \approx \left|\sum_j 1\right|^2 = N^2.$$

$\Rightarrow$ coherent emission is dramatically enhanced.